UNIT – III

ANALYTICAL TECHNIQUES

Analytical technique is a method used to determine a chemical or physical property of a chemical substance, chemical element, or mixture. There is a wide variety of techniques used for analysis. Eg: Conductometry, Potentiometry etc.

Conductometry

Conductometry is an electrochemical method of analysis based on measuring the resistance of an electrolytic solution.

Theory: Electrolytic conductivity is a measure of the ability of a solution to carry an electric current. Electrolytic solutions conduct current by the migration of ions under the influence of an electric field. Like a metallic conductor, they obey Ohm's law, **E = IR**

where, E is the applied potential, I is the current and R is the resistance, which is the measure of hindrance caused for the flow of current under the potential applied.

The reciprocal of the resistance is called the conductance, **C = 1/R**.

The resistance of any conductor is directly proportional to the length (l) and inversely proportional to the area of cross section (a) of the conductor.

R = ρ (*l*/*a*) Where ρ is the resistivity or specific resistance of the conductor.

Therefore, conductance, $C = {}^{1}/_{R} = \frac{1}{\rho}\left(\frac{a}{l}\right)$

$$C = \kappa\,(a/l)$$

where, κ is the specific conductance or conductivity of the electrolyte solution.

When $a = 1$, $l = 1$, C = κ or s**pecific conductance** is the conductance of an electrolyte solution kept between two electrodes of 1 m^2 cross sectional area at 1 m apart. It is the conductance of a meter cube of the solution. The unit of κ in SI units is Sm^{-1}.

In electrolytic solutions, the ion concentration is an important variable. It is usual to relate the electrolyte conductance in terms of equivalent conductance or molar conductance.

Equivalent conductance is defined as the conductance of a solution containing one-gram equivalent weight of the electrolyte.

Molar conductance is defined as the conductance of a solution containing one mole of an electrolyte. Conductance of an electrolyte is directly proportional to (i) mobility of ions and (ii) number of ions present in unit volume. Measurement of conductance can be employed to determine the end point in acid-base titrations. In conductometric titrations, there is a change in conductance of the solution at the equivalence point. The principle underlying conductometric titrations is the substitution of ions of one mobility by ions of another mobility of ions. The equivalence point is determined graphically by plotting conductance (ordinate) against titre values (abscissa).

Instrumentation:

It consists of a conductivity cell made up of two platinum electrodes and a conductivity meter (Fig. 1). The two electrodes have unit area of cross section and are placed unit distance apart.

The analyte is taken in the beaker and titrant is added from micro burette. Magnetic stirrer is used to stir the solution.

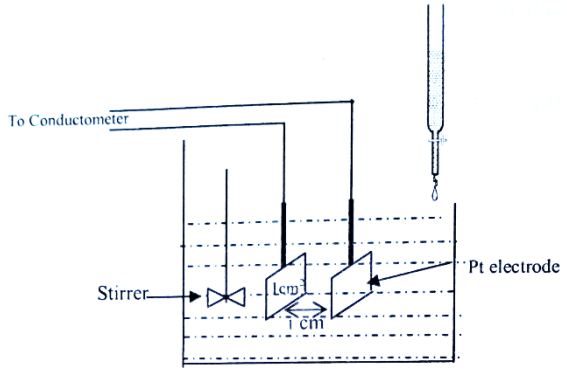


Fig. 1 Conductometric titration unit

i) Strong acid vs strong base: (HCl with NaOH)

$$\mathbf{HCl\ (H^+ + Cl^-) + NaOH\ (Na^+ + OH^-) \rightarrow NaCl\ (Na^+ + Cl^-) + H_2O}$$

In the case of strong acid and strong base, the conductance first falls, due to the replacement highly mobile H^+ ion by less mobile Na^+ ions. After the equivalence point, the conductance rapidly rises with further addition of strong base and is due to increase in the concentration of the high mobile OH^- ions (Fig. 2).

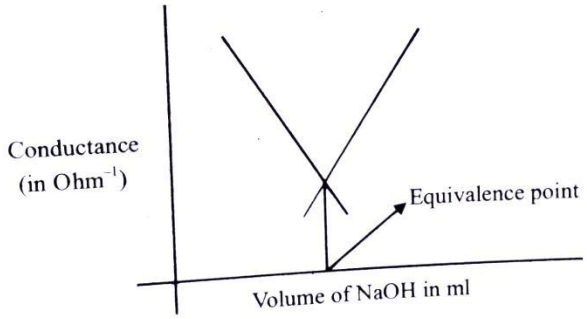


Fig. 2 Conductometric titration curve of strong acid (HCl) with NaOH

(ii) Mixture of acids vs strong base (HCl + CH_3COOH with NaOH)

$$\mathbf{HCl\ (H^+ + Cl^-) + NaOH\ (Na^+ + OH^-) \rightarrow NaCl\ (Na^+ + Cl^-) + H_2O}$$
$$\mathbf{CH_3COOH\ (CH_3COO^- + H^+) + NaOH\ (Na^+ + OH^-) \rightarrow CH_3COONa\ (CH_3COO^- + Na^+) + H_2O}$$

Initially, strong acid reacts with strong base result in steep decrease in conductance due to the replacement of highly mobile H^+ ion by less mobile Na^+ ions. After complete neutralization of strong acid, weak acid undergo reaction with strong base. Compared to partially dissociated CH_3COOH in aqueous solution, the salt CH_3COONa is highly ionized and thus conductance increases gradually with the neutralization of weak acid. On complete neutralization of the acid, addition of base lead to an increase in the number of OH^- ions and hence conductance increases sharply (Fig. 3).

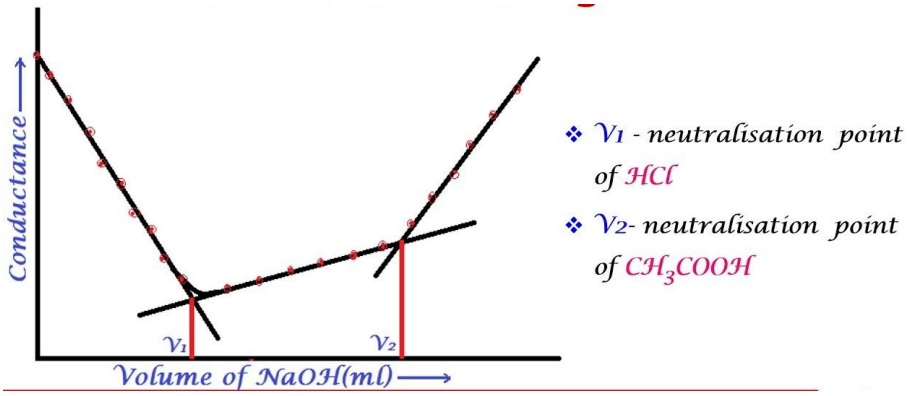


Fig. 3 Conductometric titration curve of acid mixture ($HCl+CH_3COOH$) with NaOH

Advantages:

(i) Mixtures of acids can be titrated. (ii) Accurate in dilute solutions. (iii) Very weak acids which can not be titrated potentiometrically, can be titrated conductometrically.

Potentiometry

Potentiometry is an instrumental method of analysis used to determine the concentration of a substance in a solution by measuring the potential difference between reference and indicator electrodes dipped in the solution and combined to form a galvanic cell.

Theory: The potential of an electrode is given by Nernst equation,

$$\mathbf{E = E^0 + \frac{0.0591}{n} \log [M^{n+}]}$$

where, E is the electrode potential, E^0 is the standard electrode potential, depends on the concentration of the ion $[M^{n+}]$. E can be measured by combining the electrode with a reference electrode and measuring the emf of the cell. The concentration can be calculated, provided E^0 of the electrode is known.

Instrumentation:

The potentiometer consists of a reference electrode, an indicator electrode and a potential measuring device. The indicator electrode responds rapidly to the changes in the concentration of analyte i.e the solution under study. A simple arrangement of potentiometric titration is depicted in Fig. 4, where A is the reference electrode (saturated calomel electrode), B is the indicator electrode (Platinum) and C is a mechanical stirrer. The solution to be titrated is taken in the beaker. A known volume of analyte is taken in the beaker and its potential is determined. The titrant is added in increments of 0.5 cm^3 and emf is measured each time. At the approach of equivalence point, the emf tends to increase rapidly. Few readings are taken beyond the end point. Thus, changes in potential at different volumes of titrant are recorded.

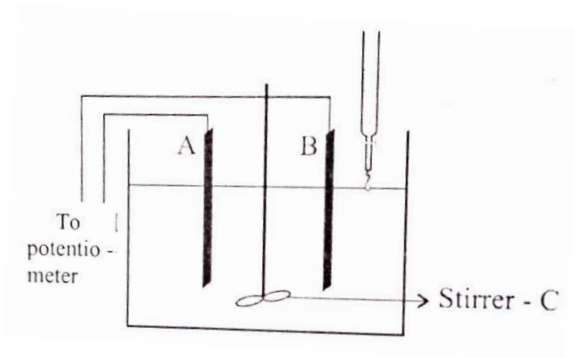


Fig. 4 Potentiometric titration unit

Applications:

Estimation of Fe: Oxidation reduction titration can conveniently be carried out as potentiometric titrations. For example, in the titration of Mohr's salt solution with potassium dichromate solution, calomel electrode is used as the reference and an inert platinum foil is used to pick up the potential of indicator electrode. Pt electrode is sensitive to oxidation-reduction reactions and its potential is sensitive to the redox species present in the solution. Before the titration is started, the solution contains only ferrous ions in the solution. When a small volume of the dichromate solution is added, equivalent small quantity of Fe^{2+} ions are converted into Fe^{3+} ions. In the process, the Cr^{6+} ion in dichromate is reduced to Cr^{3+} ion.

$$3\,Fe^{2+} + Cr^{6+} \longrightarrow 3\,Fe^{3+} + Cr^{3+}$$

The presence of Fe^{2+} and Fe^{3+} ions in the solution gives rise to an oxidation reduction electrode, the potential of which can be picked up by the platinum electrode dipped in the solution (Pt / Fe^{3+}, Fe^{2+}). The electrode potential of the so formed electrode is given by,

$$\mathbf{E = E^0 - 0.0591 \log \{[Fe^{2+}] / [Fe^{3+}]\} \text{ at } 298K}$$

The electrode potential of the indicator electrode depends upon the ratio of the concentrations of reduced and oxidized species in the solution. As the titration proceeds, the concentration of Fe^{3+} goes on increasing and that of Fe^{2+} goes on decreasing. As a result, the ratio in the expression for electrode potential goes on increasing and the increase in the value of the ratio becomes very large near the end point.

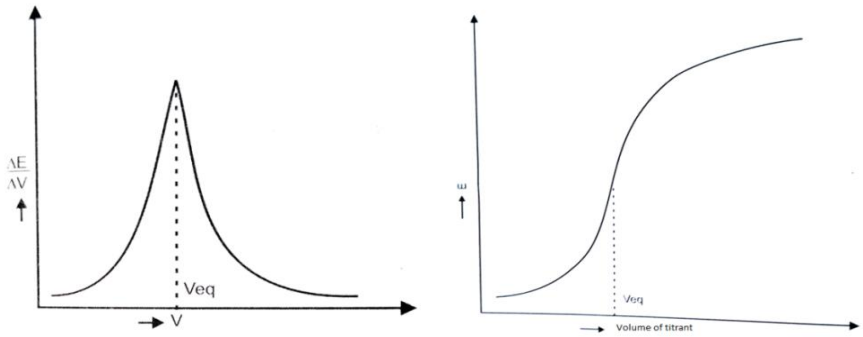


Fig. 5 Potentiometric titration curves

At the equivalence point, all the Fe^{2+} ions are converted into Fe^{3+} ions, the Pt / Fe^{3+}, Fe^{2+} electrode ceases to exist. But addition of a slight excess of dichromate solution introduces Cr^{6+} ions into the solution, which along with the Cr^{3+} ions in the solution (formed during the oxidation of Fe^{2+}) form a new oxidation reduction electrode, Pt / Cr^{6+}, Cr^{3+}. This change over of indicator electrode at the end point also contributes to the large increase in potential at the end point, as E^0 of the two electrodes differ by a large value. After the end point, therefore it is Pt / Cr^{6+}, Cr^{3+} acting as the indicator electrode.

Determination of pKa value of a weak acid:

Principle: A weak acid, like acetic acid dissociates in its aqueous solution only slightly, as follows:

$$CH_3COOH + H_2O \rightleftharpoons H_3O^+ + CH_3\,COO^-$$

The equilibrium constant of dissociation of the acid known as dissociation constant of the acid, K_a is given by

$$K_a = \frac{[H_3O^+][CH_3COO^-]}{[CH_3COOH]}$$

By taking log on both sides and simplification, we obtain Henderson's equation

$$pH = pKa + log\frac{[CH_3COO^-]}{[CH_3COOH]}$$

When acetic acid (titrate) is titrated with NaOH solution (titrant), the salt sodium acetate is formed (CH_3COONa) and pH of the titrated solution increases in accordance with the above equation. As the titration progresses, since the concentration of the salt [CH_3COO^-] increases and concentration of unreacted acetic acid [CH_3COOH] decreases, when the acid is half-titrated, [CH_3COO^-] = [CH_3COOH] and hence pH = pKa.

That is, pKa of the acid is pH of the half-titrated acid solution or pH of the titrated solution at half way to the equivalence point or end-point.

A graph of ΔpH / ΔV against volume of NaOH is plotted and equivalence point is determined. Further, using the graph of pH against volume of NaOH, pKa value of the given acid can be determined (Fig. 6).

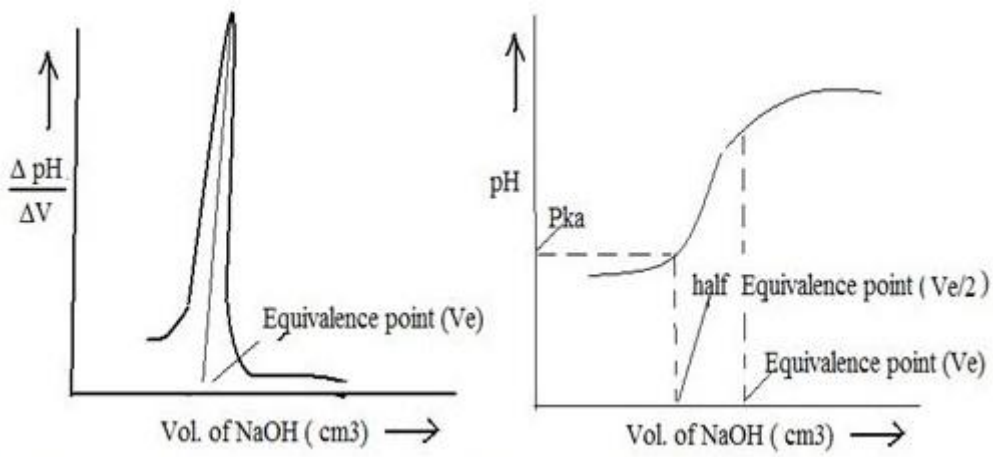


Fig. 6. pH titration curves

Colorimetry

In colorimetry, variation of colour change with concentration of analyte is determined. Colorimetry is based on emission, absorption, reflection or scattering of light by the analyte. Intensity of light emitted depends upon the concentration of the analyte. Hence, colorimetery technique is used to determine the concentration of coloured chemical species in solution. They are based on measurement of absorbance of transmittance of light of particular wavelength.

Principle: Colorimetry is governed by the Beer-Lambert's law, which states that "when a monochromatic light passes through a solution, the rate of decrease in intensity with thickness of the medium is proportional to the intensity of incident radiation as well as to the concentration of the solution."

Mathematically, $$\frac{dI}{dt} = KIc$$

where I is the intensity of the incident light, t is the thickness of the medium, c is the concentration of the medium and K is a proportionality factor.

On simplification, $$\log I_o/I_t = \in ct$$

where A is called absorbance (or optical density) of the light absorbing transparent solution. For a given substance $\in$ is constant at a given wavelength and also for the same glass cell tube the path length 't' is a constant. Therefore, $A \propto C$. Hence, a plot of absorbance against concentration of the solution gives a linear graph passing through the origin, there by verifying Beer-Lambert's law.

Instrumentation: The essential parts of a photoelectric colorimeter are a light source, a light filter, a container for the solution (cuvette), a detector to receive the transmitted light which is then amplified by an amplifier (Fig. 7). The function of the filter is to isolate the desired spectral region by filtering off the undesired radiations. Optical filters consist of either thin films of gelatin containing different dyes or colored glass. The filter allows radiations of definite wavelength range to pass through it and reach the sample.

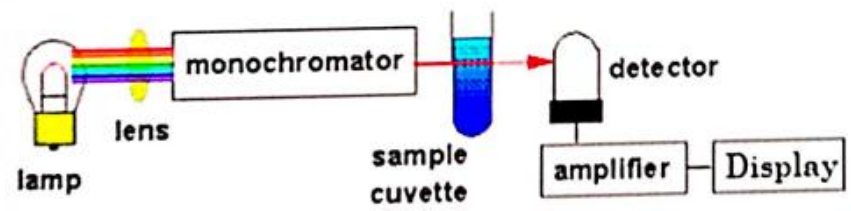


Fig. 7 Major components of a colorimeter

Colorimetric determination of Iron:

Dilute solution containing ferric Fe^{3+} ions is almost colourless. When thiocyanate is added, deep red $[Fe(SCN)_6]^{3-}$ complex is formed. In colorimetric determination, a large excess of thiocyanate should be used, since this increases the intensity and also stability. Strong acids (HCl or HNO_3) should be present to suppress the hydrolysis of Fe^{3+} ions. H_2SO_4 is not suitable, because SO_4^{2-} ions have a tendency to form complex ion with Fe^{3+} ions. A series of standard solution of iron salt and test solution is treated with dil HNO_3 (5mL) and 20% KCNS (5mL) solution and is diluted to a definite volume. The absorbance of each of these solutions

and test solution is measured at 500 nm since the complex shows maximum absorbance at this wavelength. The absorbance values are plotted against concentration (mg / mL) of iron to get a calibration curve and concentration of iron in unknown solution is determined from the graph (Fig. 8).

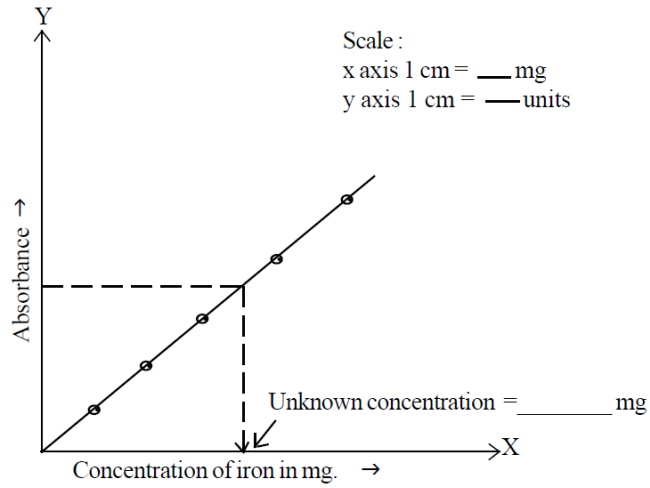


Fig. 8 Plot of concentration of iron vs absorbance

Estimation of weight of copper in industrial effluent by colorimetry

The ammonium ion reacts with cupric ion to give the complex $[Cu(NH_3)_4]^{2+}$ which has an intense blue color. The quantity of light absorbed by these colored ions can be measured using a colorimeter.

$$Cu^{2+} + 4NH_3 \rightarrow [Cu(NH_3)_4]^{2+}$$

Series of solutions containing different known concentration of the cupric ions are prepared and they are treated with ammonia solution to get cuprammonium complex. Their absorbance is measured in colorimeter at maximum wavelength of 620 nm. The absorbance values are plotted against concentration of copper (mg / mL) to get a calibration curve and concentration of copper in unknown copper in electroplating effluent is determined from the graph (Fig. 9).

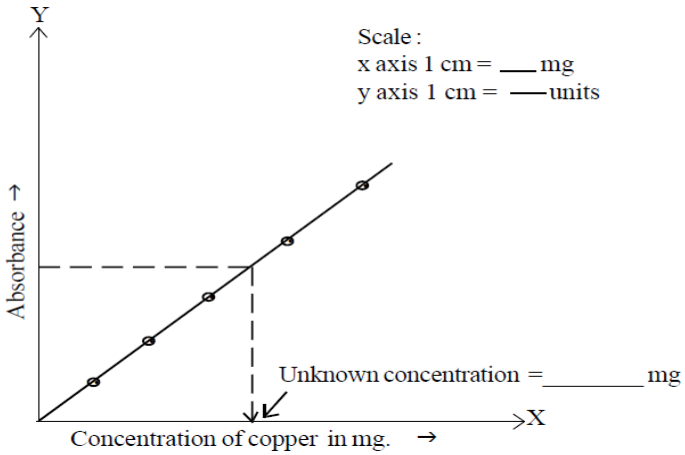


Fig. 9 Plot of concentration of copper vs absorbance

Advantages of instrumental method of analysis over chemical analysis:
(i) Chemical methods are time consuming procedures. It requires large time for analysis but instrumental methods are much faster. (ii) Chemical methods require large amount of the analyte (sample) but instrumental method requires very small amount of sample like ppm level. (iii) They are wide applications in industries. (iv) The analytical process can be automated.

Disadvantages: The instruments are expensive. (ii) An initial or continuous calibration is required using a sample of material of known composition (iii) The concentration range is limited. (iv) Specialized training is needed for the operation of certain sophisticated instruments.

Sensors
A device which gives real time analytical information about a test sample is known as sensor. Test sample is known as analyte. Sensors are designed to satisfy the growing demand for development of new instruments, techniques, and procedures, which makes analysis convenient, fast, reliable, and accurate.

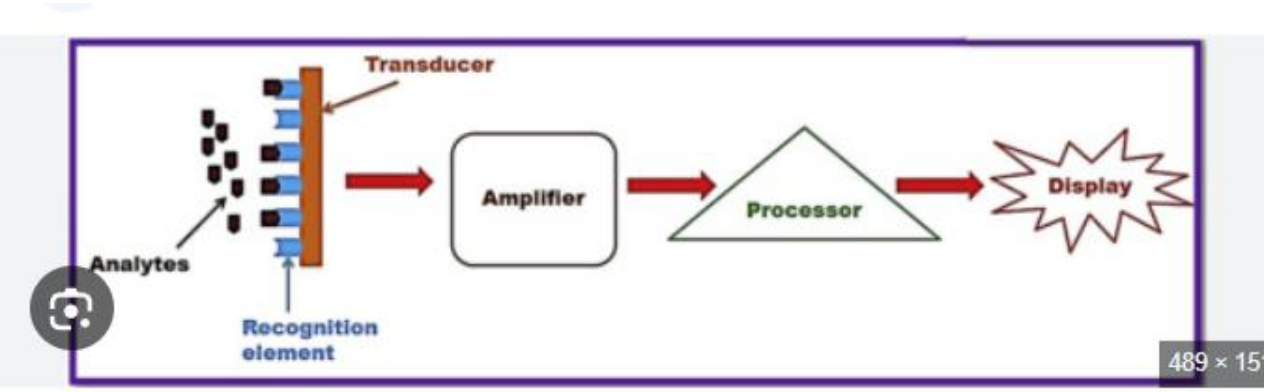


Fig. 10 Components of Sensors

Generally, a sensor is composed of three major components: Receptors, transducer and display (Fig. 10).
Receptor: Receptor detects changes in the environment. It is capable of interacting with analyte specifically and selectively. It produces a signal corresponding to interaction in the form of change in potential, conductivity, current, mass, heat.
Transducer: During the interaction of the sensing element with analyte, certain physical or chemical properties of the sensing element changes proportionately to the analyte concentration. Transducer is physicochemical device that converts chemical interactions into electrical signals that could be detected and displayed easily. Most often, sensing unit and transducer are packed together in a single unit. In case of electrochemical sensors, the electrode acts as the transducer.
Electrical signal and display: Electronic system analyses the given signal by the transducer, performs signal amplification and converts the signal from analogue to digital form. These

amplified signals are then displayed. Signals can be displayed in various forms such as numeric value, graphs images etc.,

Electro-chemical gas sensor:

Electrochemical sensors react with the measured gas and generate an electrical signal proportional to the gas concentration. Most electrochemical gas sensors are current sensors, producing a current that is linearly proportional to the gas concentration. Hence, electrochemical gas sensor is used in monitoring of concentration of gaseous analytes. They are used mainly to monitor the concentration of air pollutants, detection of leakage of chemicals in industries, and in defence, military and space applications.

Working principle: Electrochemical gas sensors is based on the principles of electrochemistry. The working (sensing) electrode and auxillary electrode are separated by electrolyte and connected to the external circuit with a small resistance. When the gas diffuses into the sensor, it reacts to generate an electric current. The current generated is proportional to the gas concentration.

Initially, gas passes through a diaphragm that prevents condensation, and this diaphragm also has a dust-proof effect. The gas molecules then diffuse through the filter and reach the working electrode. Here, the gas is oxidized or reduced, and this electrochemical reaction causes a change in the current flowing through the external circuit.

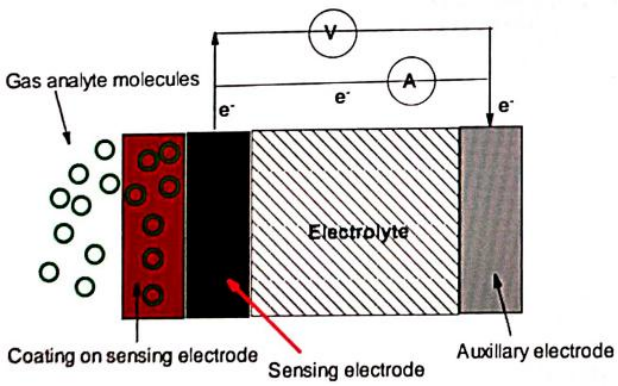


Fig. 11 Construction of an electrochemical gas sensor

Detection of oxides of sulphur

Sulfur oxides (SO_2, SO_3) released from thermal power stations and other industrial plants, are gases harmful to the environment. Sulphur dioxide is one of the main substances that pollute the air. The emission of SO_2 from industrial flue gases needs to be monitored and controlled on a regular basis.

Gas sensors detect chemicals in the gas phase. A typical SO_2 sensor consists of a three-electrode system (i) Generally, Au is used as the working (sensing) electrode (ii) Pt is used as the counter electrode and (iii) Ag/AgCl is taken as reference electrode. The chamber is filled with sulphuric acid which acts as electrolyte. A gas-diffusion membrane (Nafion or Teflon) separates the electrochemical sensing chamber from the external environment (Fig. 12).

Working: Initially, the SO_2 gas molecules pass through a small capillary-type opening and then diffuses through the hydrophobic barrier, and eventually reach the sensing electrode surface. Here, the SO_2 molecules undergo oxidation, generating electrons that flow through the external circuit and reach the counter electrode where they reduce the H^+ ions to H_2 gas. The acidic electrolyte promotes the oxidation of SO_2 and reduction of H^+ ions. The reference electrode does not directly participate in the reactions but helps in maintaining a constant reference potential throughout the detection.

Anode : $SO_2 + 2H_2O \rightarrow SO_4^{2-} + 4H^+ + 2e^-$

Cathode : $2H^+ + 2e^- \rightarrow H_2$

Net reaction : $SO_2 + 2H_2O \rightarrow SO_4^{2-} + 2H^+ + H_2$

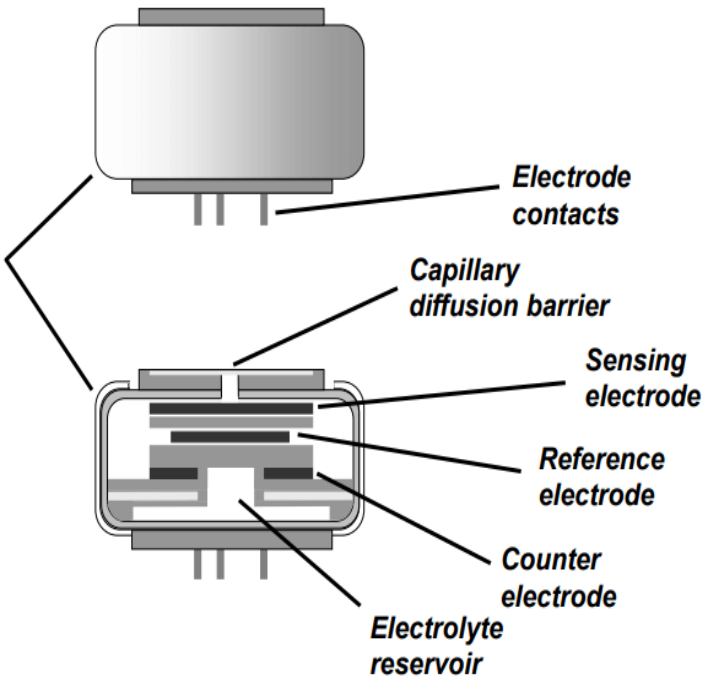


Fig. 12 Electrochemical SO_2 gas sensor

The electrochemical SO_2 sensor has its own advantages like continuous quantitative measurement; high selectivity and sensitivity (detection in ppm level); easy operation and fast response (< 20s); simple construction and low price.

Applications: (i) To detect SO_2 emission level in thermal power plants and chemical industries (ii) For monitoring air quality

Detection of diclofenac

Diclofenac, with chemical name 2-[(2,6-dichlorophenyl) amino] phenyl acetic acid, is one of the most frequently prescribed non-steroidal anti-inflammatory drugs (NSAID) with antipyretic and analgesic effects. It is safe in prescribed dose, but may cause adverse effects at higher doses. Due to wide usage and poor biodegradability, it can have serious effects on the ecosystem.

Electrochemical sensor can be used to detect diclofenac in lower concentrations. The electrode system consists of

Sensing electrode: Carbon paste electrode with multi walled carbon nanotube (MWCNT) and gold nanoparticles

Reference electrode: Ag/AgCl electrode

Counter electrode: Pt wire

The electrodes are screen printed on an insulating PVC substrate (Fig. 13). The electrochemical oxidation of DCF takes place at a neutral pH of 7.0. When a drop of the sample containing diclofenac is placed on the sensor, oxidation of diclofenac occurs at the carbon electrode to release an electron. As a result, a radical intermediate is formed which immediately undergoes hydrolysis to form 2,6-dichloro aniline and 2-hydroxyphenylacetate. Reactions on the electrode cause the current to flow. The intensity of this current is a function of the number of oxidized molecules.

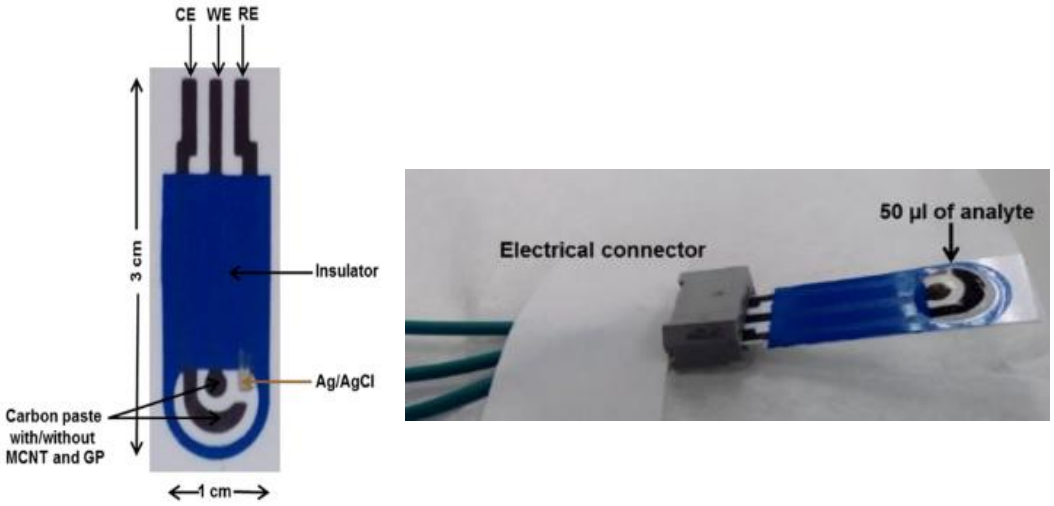
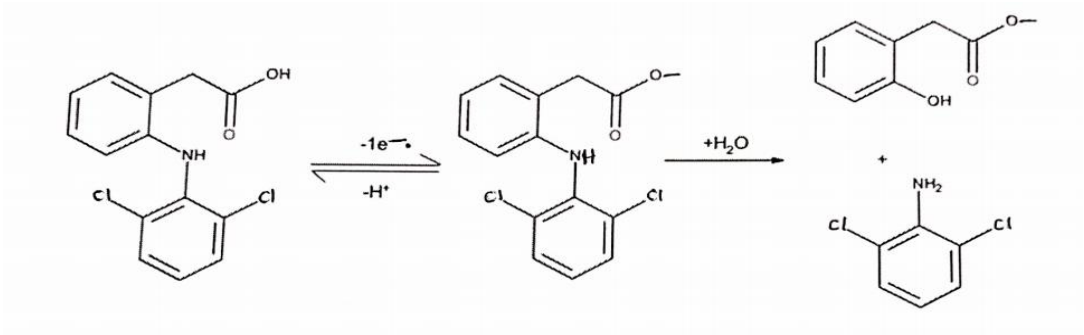


Fig. 13 Screen printed sensor for Diclofenac



Disposable Sensors:

Disposable sensors find major applications in spot analysis and can be disposed immediately after use. They should be economical, easy to use with small quantity of the sample. These sensors are made of a special type of paper, over which receptor and electrodes are coated. They are coated using screen printing technology (Fig. 14).

Several types of disposable biosensors have been developed for continuous monitoring and on spot analysis of biomolecules such as glucose, ascorbic acid, uric acid, creatinine, lactic acid and dopamine in human serum samples.

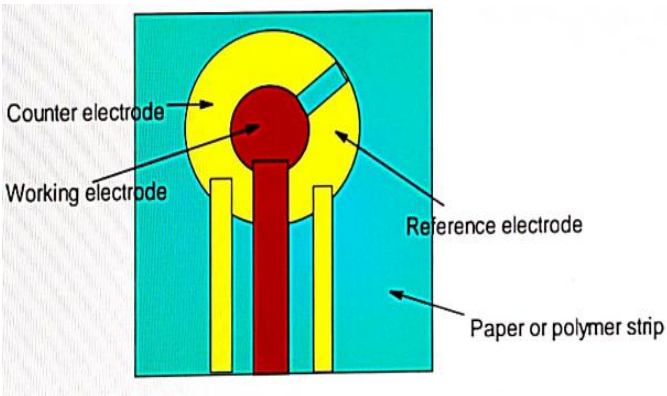


Fig. 14 Screen printed paper strip disposable sensor

Disposable sensors for glucose

Disposable glucose sensors are widely used in devices like blood glucose meters to monitor blood sugar levels in people with diabetes. These sensors are typically electrochemical, utilizing specific biochemical reactions to detect the presence and concentration of glucose. The sensing electrode (Au) and counter electrode (Pt foil) are screen printed on a disposable substrate such as paper. The sensing electrode is often modified with an enzyme such as glucose oxidase which is immobolized on the sensing electrode surface. An insulating layer separates the two electrodes and creates a capillary channel to draw in the sample.

When a drop of blood containing glucose comes into contact with the sensor, glucose molecules diffuse to the enzyme layer. At the surface of the working electrode, glucose oxidase (GOx) catalyzes the oxidation of glucose to gluconic acid. The generated current is measured and processed by the in-built transducer and converted in to glucose concentration (mg / ml) which gets displayed on the digital screen of the glucose meter.

$$CH_2OH-(CHOH)_4-CHO + H_2O \xrightarrow{GO_X} CH_2OH-(CHOH)_4-COOH + 2H^+ \; 2e^-$$

Advantages of disposable electrodes:

1. Easy to use
2. Small quantity of analyte is sufficient
3. Can be disposed immediately after use
4. More accurate and rapid

Applications of disposable electrodes:

1. Medical and pharmaceutical applications
2. Detection of heavy metals and organic pollutants in environment
3. Detection of trace amount of radioactive elements
4. Water and soil analysis

MEMORY DEVICES

An electronic memory device is a form of semiconductor storage device which is fast in response and compact in size, and information can be read and written when coupled with a central processing unit (CPU, a processor).

The basic goal of a memory device is to provide a means for storing and accessing **binary digital data** sequences of "1s" and "0s", as one of the core functions (primary storage) of modern computers. In conventional silicon-based electronic memory, data are stored based on the amount of charge stored in the memory cells.

Requirements of materials to exhibit memory effect

- The individual memory cells, must possess at least two stable states. In silicon- based electronic memory devices used today, data are stored based on the amount of charge stored in the memory cells. Memory cells exhibit two states ("ON' and OFF which are coded as "1" and "0").
- These states must be stable for a period appropriate for the data storage.

In addition to above points, parameters of importance to the performance of a memory cell include switching time (write and erase), ON/OFF current ratio (or memory window), read cycles, and retention ability.

The switching time influences the rate of writing and accessing the stored information. The ON/OFF current ratio defines the control of the misreading rate during device operation. Higher value of ON/OFF value being essential for the device to function with minimal misreading error. The number of read cycles and retention ability are related to the stability and reliability of the memory devices.

For practical applications, other factors, such as power consumption and cost, structural simplicity and packing density, as well as mechanical stiffness and flexibility, are of equal importance when designing and fabricating new memory devices.

Classification of electronic memory devices:

With their respective ability to amplify electronic signals, to store charges, and to produce proportional electric currents, electronic memory devices can be constructed from four primary categories of device structures namely transistor-type, capacitor-type, resistor-type and charge transfer-type.

1. Transistor-type electronic memory device

In order to store data in a memory device, it must have minimum two distinct electronic states. These states are assigned as "0" and "1" or "ON" and "OFF" respectively. A transistor-type semiconducting electronic memory device consists of at least three terminals for connecting the electronic circuit. They are emitter, collector and base (Fig. 1).

A typical transistor will have PNP or NPN junction. A transistor can act as a switch or gate for electronic signals, opening and closing an electronic gate many times per second. It ensures the circuit is ON if the current is flowing and switched OFF if it isn't. Each transistor represents one bit of information which is 0 or 1. In this electronic circuit, "0" and "1" corresponds to the "discharged" and "charged" states, respectively.

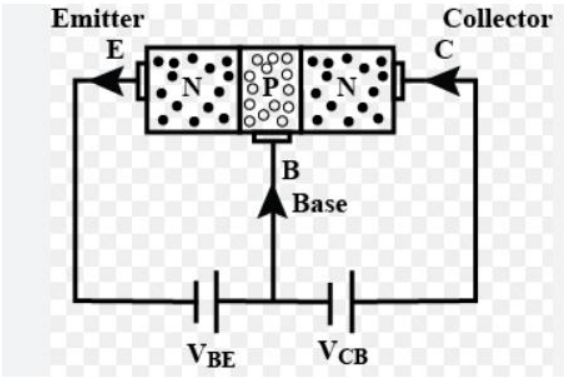


Fig. 1 Transistor-type memory device

2. Capacitor- type electronic memory device

Capacitors have two parallel plate electrodes separated by an insulating material known as a dielectric (Fig. 2). Charges are stored in these electrodes under an applied electric field. Bistable states of capacitor are based on the amount of charge stored in the cell. Each capacitor represents one bit of information. When capacitor is charged, it represents 1 and when discharged, it represents 0.

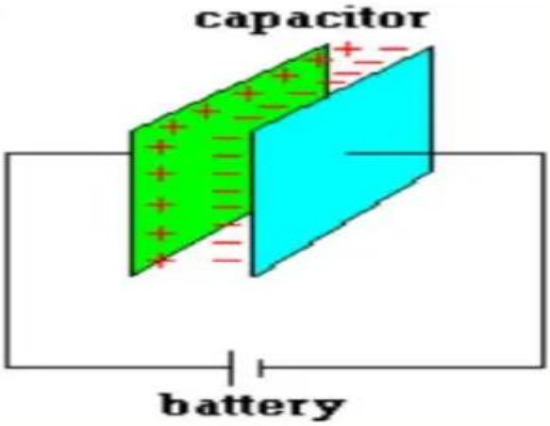


Fig. 2 Capacitor-type memory device

3. Resistor-type electronic memory device

This type of memory device uses switchable resistive materials to store data. In other words, the resistive devices are made up of metal-insulator-metal structure. Middle layer acts as an active switching medium (Fig. 3). It is based on the change of the electrical resistivity of materials in response to an applied voltage (electric field). The low resistance state represents 1 and high resistance state represents 0 state.

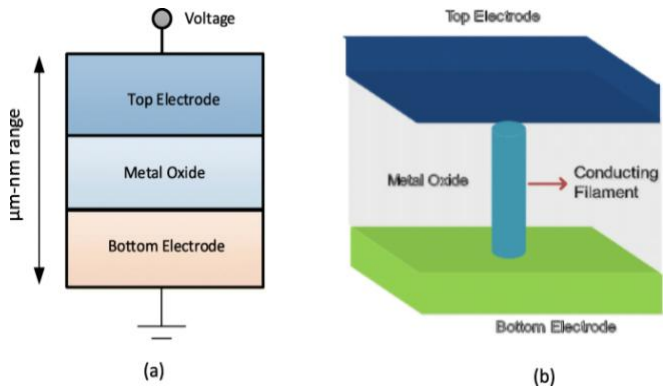


Fig. 3 Resistor-type memory device

4. Charge transfer-type electronic memory device

This type of electronic device is based on the Charge Transfer Effects of a charge transfer complex. A charge transfer (CT) complex consists of two parts, one electron donor and other an electron acceptor. It is also called as a donor-acceptor (D-A) complex. The conductivity of a CT complex is dependent on the ionic binding between the D-A components. In CT complex, a partial transfer of charges occurs from donor part to the acceptor part. This results in difference in conductivity. CT complexes exhibit bistable states due to difference in conductivity.

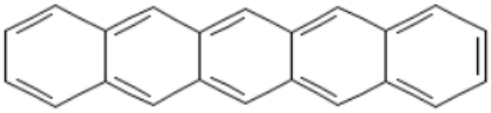
Types of Organic Memory Devices

Organic memory devices have shown the potential to be the new data storage technologies. These materials exhibit two electrical stable states known as electrical bi-stability. They shift from one state (ON state) to other state (OFF state) when an external electric field is applied. In these devices, information is stored and retrieved by encoding these two states. Following are the different classes of materials which can exhibit bistable states and are used in organic memory devices.

1. Organic Molecules: There are some organic molecules which show bistable or multi stable states when external field is applied. When a threshold voltage is applied, they are capable to undergo a reversible transition from the OFF state to the ON state. All these materials can be used in organic electronic memory devices. Few of them are mentioned below:

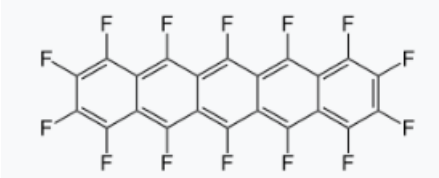
E.g. Acene derivatives: Acenes are the polycyclic aromatic compounds consisting of linearly fused benzene rings. They are the very first molecules to be recognized as organic memory devices because of their high charge carrier mobility. An example for acenes is pentacene.

Pentacene is a linearly-fused aromatic compound with five benzene rings. It can be obtained in crystal and thin film form. Both forms exhibit a very good hole mobility. Pentacene molecule is reported to exhibit a hole mobility upto 35 cm^2/Volt.sec and hence it behaves as a p-type semiconductor.



Pentacene molecule

When all the hydrogen atoms of pentacene are replaced by fluorine atoms the resulting molecule is perfluoropentacene. Strongly electron withdrawing nature of fluorine atoms convert this molecule in to n-type semiconductor.

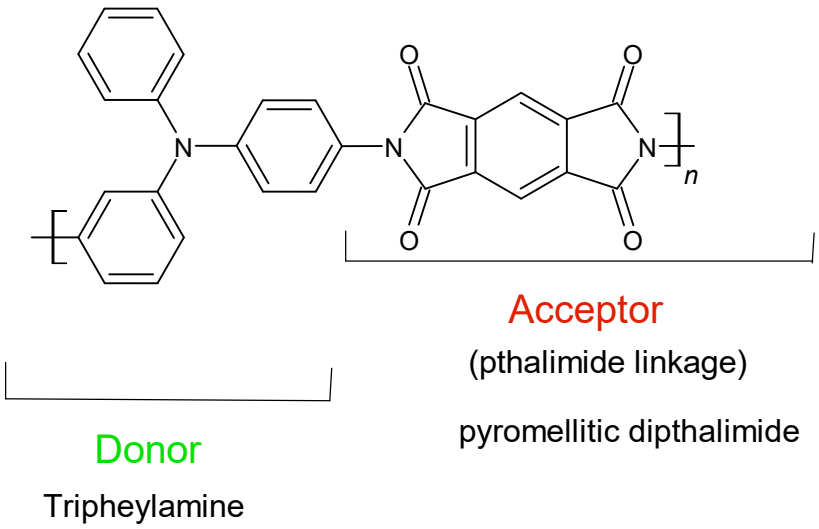


Perfluoropentacene molecule

Pentacene and Perfluoropentacene, both have similar structure and similar crystal packing but former behaves as p-type semiconductor and latter behaves as an n-type semiconductor. Using them together in heterojunctions enables ambipolar devices, where both electrons and holes can be transported efficiently. Pentacene and perfluoro-pentacene are mainly used in transistor type and capacitor type electronic memory devices for non-volatile data storage.

2. Polymer Molecules: There are different classes of polymers which exhibit memory effect and are used in electronic memory devices.

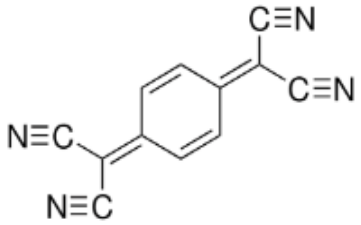
E.g. Functional polyimides - Functional polyimides (PIs) are one of the most commonly used polymeric materials for organic electrical memory applications. They have high thermal stability and mechanical strength and can be easily processed from solution. In functional PIs, triphenylamine acts as an electron donor and phthalimide acts as the electron acceptor, thereby forming a Donor-Acceptor structure. They exhibit two stable charge states under applied electric field. These states arise due to transfer of electrons from donor to acceptor. This bi-stability is used to store data in memory device.



3. Organic-Inorganic Hybrid Materials: Organic-inorganic hybrid materials are composed of organic layers containing inorganic materials. Most often, the organic molecules are electron rich or deficient molecules or polymers containing electron donors such as thiophene, fluorine, carbazole and aniline derivatives. The inorganic materials used are metal nanoparticles, semiconductor nanoparticles and inorganic quantum dots (QDs). Hybrid materials involving an inorganic metal center surrounded by an organic molecule exhibit charge transfer and are widely used in memory devices for their switchable electrical conductivity.

E.g. Cu-TCNQ charge transfer complex: Charge transfer complexes are composed of electron donor and electron acceptor moieties. In Cu-TCNQ complex, copper, with electronic configuration [Ar]$4s^1 3d^{10}$, acts as an electron donor by giving up single electron to attain stable electronic configuration of [Ar]$3d^{10}$. Cyano group of TCNQ is strongly electron withdrawing in nature and hence acts as an electron acceptor.

These devices exhibit two stable charge states which arise due to transfer of electrons from donor to acceptor under the influence of external field and this principle is used in creating non-volatile memory arrays (ability to retain data even when power is switched off).



7,7',8,8'**-t**etra**c**ya**n**o**q**uinodimethane (TCNQ)

Advantages of organic electronic memory devices

a) They can be processed easily.
b) Structure of the molecule used can be designed through chemical synthesis,
c) Device structure is very simple.
d) Dimension of the device can be decreased (miniaturized).
e) Cost of production is less.
f) Power consumption during operation is low.
g) Data storage capacity can be increased to very large value.

DISPLAY DEVICES

Liquid Crystals

The study of liquid crystals began with an observation made by Austrian botanist, Freindrich Reinitzer in 1888. He observed that solid cholesteryl benzoate on heating becomes a turbid liquid at 145°C which on further heating turns into a clear, transparent liquid at 178°C. The above changes are reversed on cooling. The changes are generally represented as

$$\underset{\text{Solid}}{\text{Cholesteryl benzoate}} \underset{}{\overset{145^0C}{\rightleftharpoons}} \underset{\text{mesophase}}{\text{Choleteryl benzoate}} \overset{178^0C}{\rightleftharpoons} \underset{\text{Liquid}}{\text{Cholesteryl benzoate}}$$

Cholesteryl benzoate is said to exist as liquid crystal between 145°C and 178°C. The first temperature at which solid changes into turbid liquid is known as transition point and second temperature at which turbid liquid changes into clear liquid is known as melting point. Thus, liquid crystal is a distinct phase observed between crystalline solid state and isotropic liquid state. The liquid crystal state is also referred to as **mesophase**. The compounds which exhibit mesophase are also called **mesogens**. Liquid crystals exhibit optical anisotropy, i.e., they possess different optical properties when light is incident in different directions.

Definition: Liquid crystals may be described as a distinct state of matter in which the degrees of molecular ordering lie intermediate between the ordered crystalline state and the completely disordered isotropic liquid state.

Positional and Orientational Order: Most liquid crystals are composed of organic molecules. In solid state, the molecules are highly ordered. Each molecule occupies a definite position in a more or less rigid arrangement and orient in a preferred direction. In liquid state, however, the molecules neither occupy specific positions nor remain oriented in a particular manner. The molecules are somewhat free to move at random and collide with one another abruptly changing their positions. In the intermediate state, i.e. liquid crystal phase, the molecules are free to move but are oriented in a particular direction.

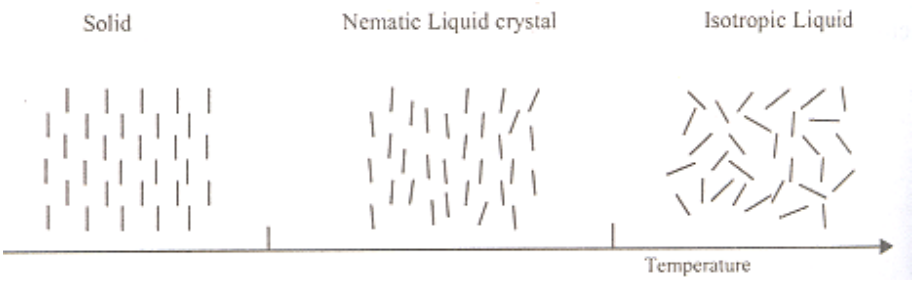


Fig. *1 Representation of solid, liquid crystal and liquid states (molecules are represented as thin lines*)

Thus, solid phases possess positional order and orientational order. Liquid phases possess neither positional nor orientational order while in liquid crystal phase some orientational order is retained though there is a loss of positional order, as shown in Fig. 1.

Director: The direction of preferred orientation in a liquid crystal is called director. It is an imaginary axis which is described as an average angle made by each molecule with the director

at any instant of time. An average of 0^0 indicates perfect orientation and can be expected in solids. An average of greater than 45^0 indicates no orientational order and is found in liquids. However, in liquid crystals a smaller average angle (between 0^0 and 45^0) with the director is observed, which indicates an orientational order to a small extent (Fig. 2).

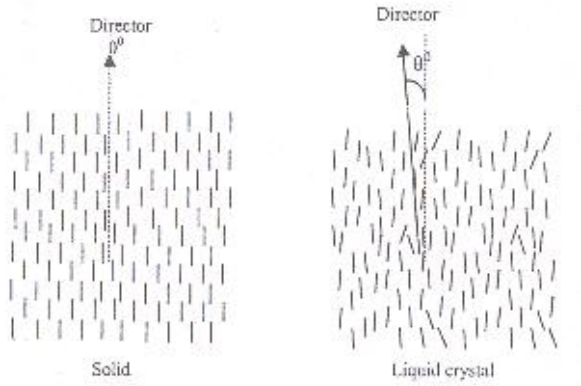


Fig. 2 A snapshot showing the orientation of the molecules in the liquid crystal phase as compared to its solid phase

Classification Of Liquid Crystals

Liquid crystals are broadly classified into two main categories, namely,

1. Thermotropic liquid crystals 2. Lyotropic liquid crystals

1. Thermotropic liquid crystals: The class of compounds that exhibit liquid crystalline behaviour on variation of temperature alone are referred to as thermotropic liquid crystals. The temperature range, at which some liquid crystals are stable, are given below:

	Solid		Liquid crystal		Liquid
Cholesteryl Benzoate		145°C		178°C	
p-Azoxyphenetole		137°C		167°C	
p-Azoxyanisole		116°C		135°C	

2. Lyotropic liquid crystals: Some compounds transform to liquid crystal phase when mixed with a solvent. They are usually obtained by mixing the compound in a solvent and increasing the concentration of compound till liquid crystal phase is observed. Such liquid crystals are called lyotropic liquid crystals. The formation of lyotropic mesophases is dependent on the concentration of either the component or the solvent. Variation of temperature also affects the formation of these mesogens.

Lyotropic mesogens are typically obtained from amphilic compounds comprising of both lyophilic (solvent attracting) and lyobhobic (solvent repelling) parts in the same molecule. In the presence of solvent, the lyophobic ends come together while the lyophilic ends get directed towards water forming micelles. The formation of micelles takes place only beyond a particular concentration of the solution called critical micelle concentration (CMC). When the concentration of the solution is increased (beyond CMC), the micelles increase in size and eventually coalesce to form liquid crystalline phase.

Example: soap (soap-water mixture) molecules

Types of Mesophases (based on Molecular Ordering)

(1) Nematic Mesophase: Nematic (Greek; nematos = thread like) liquid crystals are formed by compounds that are optically inactive. The molecules have elongated shape and are approximately parallel to one another (Fig. 3). Nematic phase is characterized by the total loss of positional order and a near normal flow behaviour similar to its liquid phase.

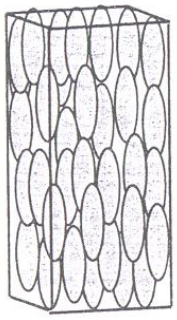
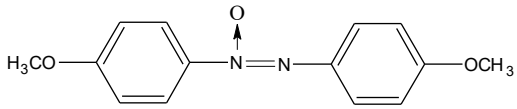


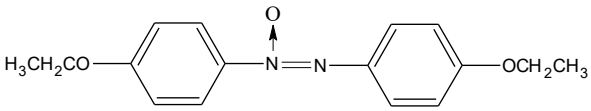
Fig. 3 Nematic liquid crystal

Examples:

(i) p-Azoxy anisole (116-135^{0}C)



(ii) p-Azoxy phenetole (137-167^{0}C)



(2) Chiral (Twisted) Nematic Mesophase: Chiral nematic liquid crystals also referred to as cholesteric liquid crystals or twisted nematic liquid crystals (TNLC), formed from optically active compounds having chiral centres. Unlike in a nematic phase where all the molecules approximately parallel to one another, in chiral nematic phase, the molecules arrange themselves in such a way that they *form a helical structure*. In this mesophase, the director is therefore not fixed in space as in a nematic phase, but rotates throughout the sample forming a helical pattern, as it changes its direction just like the motion of a nut on a screw. The distance traveled by the director as it completes one full turn is called the pitch of the liquid crystal (Fig. 4). In other words, the pitch length is the distance traveled by the director when it gets turned by 360^0. The twisted pattern repeats itself throughout the liquid crystal phase.

The most striking feature of TNLC is its strong optical activity and selective light reflection, which are attributed to the twisted structure. The twist present in chiral nematic liquid crystal imparts spectacular optical properties which are made used of as thermochromic materials. The pitch is also temperature dependent and hence cholesterics are used in thermography.

Examples: (i) Cholesteryl benzoate; (ii) Cholesteryl myristate and (iii) Cholesteryl formate etc.

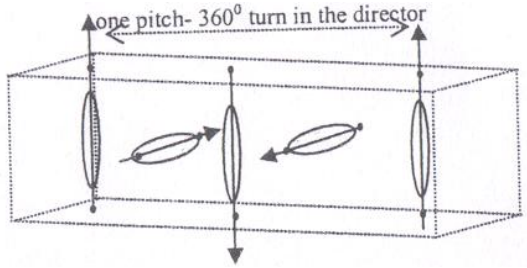
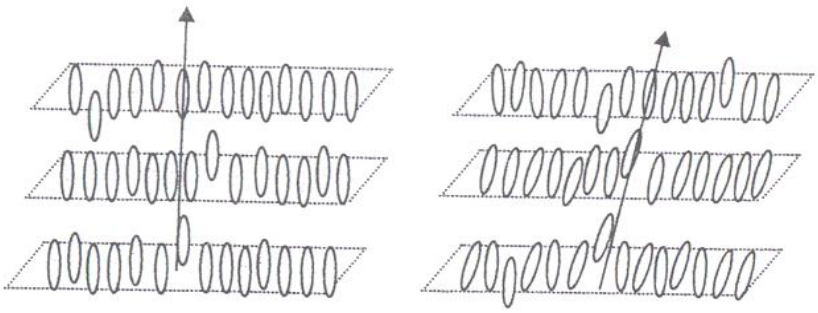


Fig. 4 Illustration of the twisted structure in a chiral nematic phase of a liquid crystal as shown by the change in the direction of the director. Arrows show the orientation of the molecules.

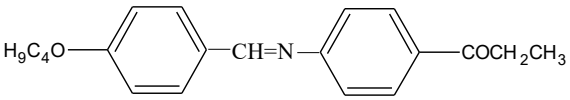
(3) Smectic Mesophase: Substances that form smectic phases are soap-like (in Greek, smectos means soap). In fact, the soft substance that is left at the bottom of a soap dish is a kind of smectic liquid crystal phase. In smectic mesophase, there is a small amount of orientational order and also a small amount of positional order. The molecules tend to point along the director and arrange themselves in layers. A snapshot would reveal that more number of molecules position in regularly spaced planes and a few molecules lie between the planes (Fig. 5). That is, any one molecule would spend more time in these planes than between the planes. Based on the orientation of the director there are many types of smectic phases. If the director is perpendicular to the planes, it is called smectic A (Fig. 5a) and smectic C if the director makes an angle other than 90^0 (Fig. 5b). In smectic B phase, the director is perpendicular to the plane with the molecules arranging themselves into a network of hexagons within the layer.



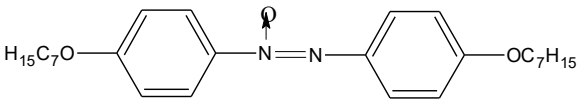
(a) (b)

Fig. 5 Schematic representation of smectic mesophase where molecules lie on regularly spaced planes (in the form of layers). (a) Smectic A where molecules lie perpendicular to the layer planes. (b) Smectic C where the molecules are tilted with respect to the layer planes.

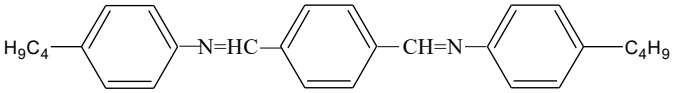
Examples:



4-n-Butyloxybenzylideneamino propiophenone (smecti A)



4,4[1]-di-n-Heptyloxyazoxybenzene (Smectic C)



Terephthalylidene-bis-4-n-butylaniline (smectic B)

(4) Discotic Mesophase: Liquid crystals formed by molecules which have disc-like or plate-like structure are referred to as discotic liquid crystals.

The simplest discotic phase is also called discotic nematic phase because there is orientational order but no positional order (Fig. 6a). There is random motion of the molecules, but on an average, the axis perpendicular to the plane of each molecule tends to orient along the director.

In the discotic columnar phase, in addition to the orientational order present in the discotic nematic phase, most of the molecules tend to position themselves in columns (Fig. 6b). The columns are arranged in a hexagonal lattice resembling a set of coins stacked as shown below (note: the coins in a stack have a great deal of positional order i.e., the coins are equidistant whereas the molecules in a columnar phase are stacked in random fashion).

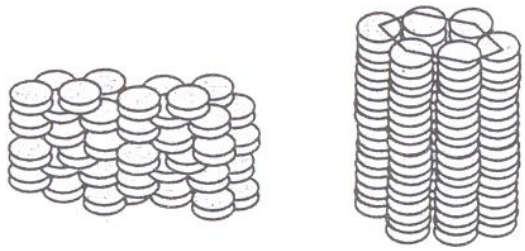
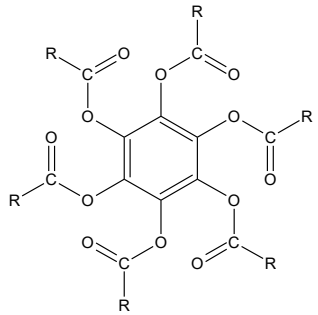


Fig. 6 Schematic representation of disc like molecules arranged in (a) Discotic nematic (b) discotic columnar liquid crystal phases

Example: Benzene-hexa-n-alkanoate



Electro-Optic Effect of Liquid Crystals

Generally, liquid crystals have rod-like molecular structure and align themselves spontaneously along the director. Nematic materials have two dielectric constants- one in the direction parallel to the director and the other perpendicular to the director. Dielectric anisotropy ($\Delta\varepsilon$) is defined as the difference between the dielectric constants parallel and perpendicular to the director. Similarly, the optical anisotropy (Δn) is defined as the refractive index parallel to the director minus the refractive index perpendicular to the director. These two properties are important for the electro-optic effects in liquid crystals.

Effect of electric field: The director in a liquid crystal is free to point in any direction. But when a film of liquid crystal is placed between two charged plates, director is forced to point along a particular direction when an electric field is applied. For example, when a film of liquid is placed between two specially treated glass sheets (such as rubbing with a velvet cloth), the

molecules close to the glass surfaces are forced to orient themselves parallel to the surfaces of the glass sheets.

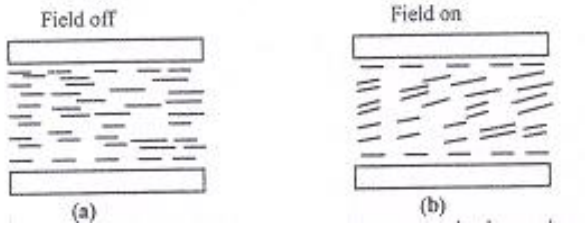


Fig. 7 Liquid crystal film kept between two treated glass plates (a) in the field off state all molecules and the director orient parallel to the surface (b) in the field on state the molecules near the surface orient parallel to the surface whereas in others it is deformed

In the absence of an electric field, (below a threshold value) the molecules at other layers are also aligned parallel to the surfaces giving a homogeneous arrangement. But when an electric field is applied perpendicular to the glass surface, molecules near the surface are aligned parallel to the surface (perpendicular to the applied field). The molecules near the centre of the liquid crystal layer (away from the surface) are free to orient themselves along the applied field. The crystal undergoes deformity (Fig. 7). This transition (deformity) is important in the operation of liquid crystal displays because the transition brings about a significant change in the optical characteristics of the liquid crystal.

Effect of light: When light is incident on two crossed polarizers, no light emerges because the light emerging from the first polarizer is completely absorbed by the second polarizer and hence appears dark. However, when a film of optically active liquid crystal (e.g. TNLC) is placed between the two polarizers, the twisted structure of TNLC acts like a wave-guide and gradually rotates the plane of polarization of light by 90^0, resulting in a bright appearance (Fig. 8a). The 90^0 twist in the cell is lost when a sufficiently strong electric field (2 - 5 V) is applied to the cell. Hence, the cell appears dark between two crossed polarizers (Fig. 8b).

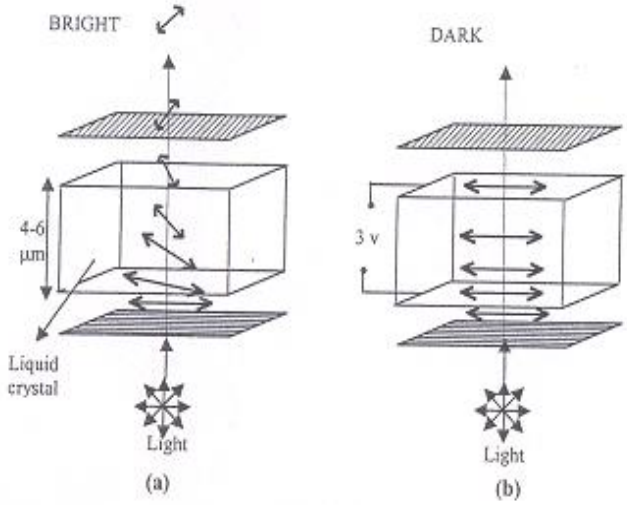


Fig. 8 Bright and dark appearance when light is passed through TNLC placed between two crossed polarizers (a) appearance is bright in the absence of electric field (b) appearance is dark in the presence of electric field.

The following conditions, however should be met to see the electro-optic effect.

(i) The plane of polarization of incident light should be parallel or perpendicular at the surface of the cell.

(ii) The product of optical anisotropy, Δn, (the difference in refractive indices parallel and perpendicular to the director) and pitch P should be greater the wavelength of the incident light. $\Delta n \times P > \lambda$.

Applications of Liquid Crystals in Numeric Display

A liquid crystal display (LCD) is a flat panel display technology that uses liquid crystals to produce images. The light is passed or blocked by changing the polarization.

LCD unit: A numeric display consists of seven segments for each digit. Light from the area of each of the seven segments of pixels is controlled independently and is used to create any one of the ten digits. Such kind of display systems are often seen in calculators and digital watches.

The top and bottom glass plates of a display have electrode patterns formed by etching a layer of indium tin oxide (ITO) using standard photolithographic process. Each of the glass plates are connected to polarizer (crossed). For colored display, a color filter with dichroic dyes (red, blue or green) is provided next to the top glass plate. A small quantity of liquid crystal mixture is filled in the cell and the edges of the top and bottom glass plates are sealed. The distance between the top and bottom glass plate is called cell thickness. It is usually in the range 4 to 8 μm to ensure that display can be switched in milliseconds. Spacers, such as silica, are used to maintain a uniform cell gap and also to prevent flowing out of the liquid crystal (Fig. 9). The bottom glass plate of each of the segments is connected to a voltage source (microcontroller) which supplies the required electric field.

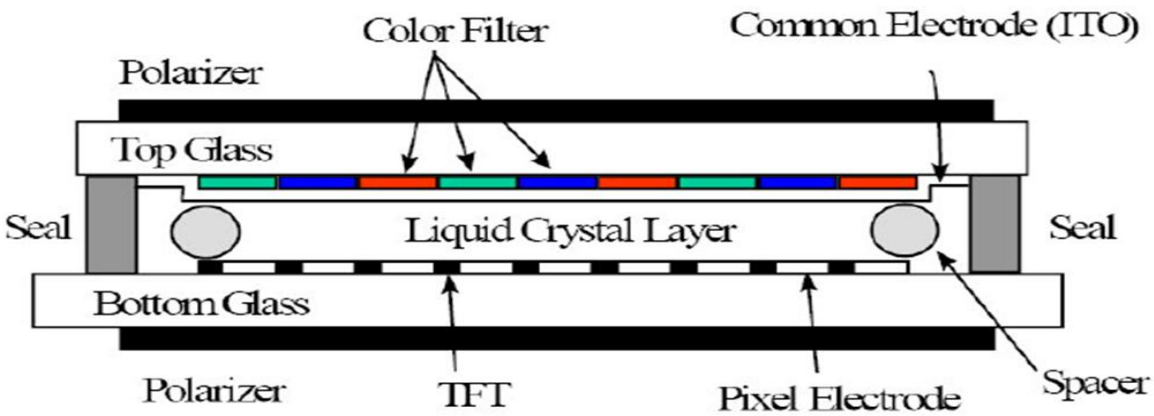


Fig. 9 A single LCD unit

Working of LCD (Numeric Display): The working principle of an LCD is based on the electro-optic effect of liquid crystals.

A LCD consists of a layer of liquid crystal material sandwiched between two transparent electrodes. The molecules adjacent to the glass will align to the polarizer, resulting in a helical structure between the two glass plates. As light strikes the bottom plate, it gets polarized. The molecules in each layer then guide the light through the display, changing the angle of polarization. When light reaches the final layer of the crystal material its plane of polarization

will have rotated 90º, and now matches the plane of the top polarizing film, hence comes out through it appearing bright.

When electric field is applied across the plates, the liquid crystal molecules deform and lose their property of rotating the light. Hence, the plane of the light no longer matches the angle of the top polarizable plate. Consequently, no light can pass through the top glass plate giving a dark appearance. In the segment arrangement, the conducting segments look dark and the other segments look clear.

As an example, Fig. 10 shows the numeric display of digit 2. On applying a voltage of about 3 volts to the back panel, the segments *A*, *B*, *D*, *E* and *G* can be switched ON. The liquid crystals in these segments turn optically inactive when subjected to the applied voltage. In the absence of electric field, segments *C* and *F* allow the transmission of light and remain bright, thereby displaying the digit 2.

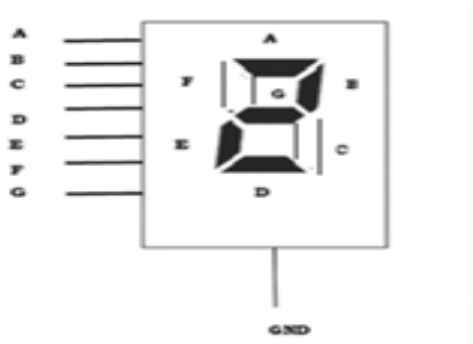


Fig. 10 LCD unit with the display of number 2

Organic Light Emitting Diodes (OLED)

OLEDs are organic electroluminescent LEDs. OLEDs can produce high quality displays with high contrasts, less power consumption and faster response time. It is considered more advanced than LCDs, which suffer from poor viewing angle. OLED devices consist of two electrodes, an anode and a cathode. Organic layers are placed between these electrodes. Here, organic compounds play a vital role in the generation of light (photons). Multiple organic layers are used, in which each layer plays an intrinsic role.

Construction: A simple OLED device has the following layers deposited sequentially over a transparent substrate (glass or plastic), through which the light gets emitted.

(i) **Anode:** Generally, anode in OLEDs is made up of transparent conductive electrodes (commonly Indium Tin Oxide – ITO). The major role of anode is to inject holes (positive charge carriers) into the organic layers when current flows.

(ii) **Conductive layer:** This layer is responsible for transporting the holes generated from the anode and hence is also called as Hole Transport Layer (HTL). Derivatives of triphenylamine are good examples of organic molecules capable of efficient hole transport.

(iii) **Emissive layer:** The emissive layer, also known as Electron Transport Layer (ETL), transports the electrons injected from the cathode. Alq3 [tris(8-hydroxyquinolinato) aluminium] facilitates the quick movement of electrons and hence, is used as ETL.

(iv) **Cathode:** Metals such as Al and Mg are employed as cathode. Their main function is to inject electrons in to the emissive layer.

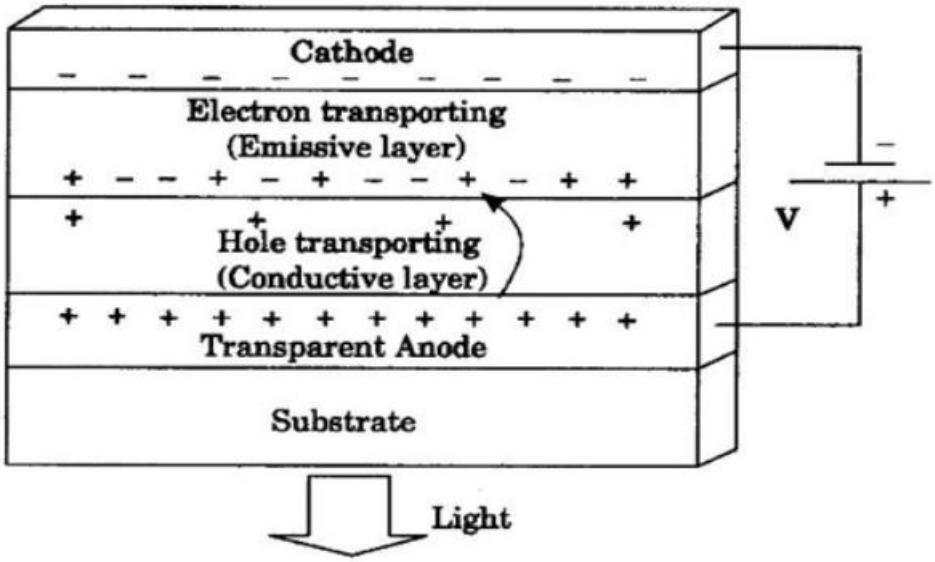


Fig. 11 Construction and working of OLED

Working: On applying voltage, the cathode injects electrons to the emissive layer while the anode withdraws an electron from the adjacent conductive layer. As a result, holes are created in the conductive layer. Gradually, the emissive layer becomes rich in electrons (negatively charged) and the conductive layer becomes rich in holes (positively charged). The holes and electrons recombine closer to the emissive layer (in organic materials, holes move faster than electrons) (Fig. 11). The bound state of electron-hole system (also called as exciton) returns to the ground state by emitting photons, resulting in the emission of visible light.

Applications of OLEDs

1. **Televisions and displays**: OLED displays are used in televisions, monitors, smartphones, and other electronic devices.
2. **Lighting**: OLEDs can also be used as a source of lighting in various applications, including automotive lighting, street lighting, and architectural lighting.
3. **Wearable devices**: The thin and flexible nature of OLEDs makes them suitable for use in wearable devices, such as smartwatches and fitness trackers.
4. **Automotive**: OLEDs can be used in automotive applications, such as dashboard displays, interior lighting, and taillights.

Quantum-Dot Light Emitting Diode (QLED)

QLED is a display technology that is based on the principle of electroluminescence produced by quantum dots. Quantum dots (QDs)are nanoscale semiconductor particles (2-10 nm) exhibiting size-tunable emission (quantum confinement effect) i.e. they are capable of emitting light at different wavelengths (colors) depending on their size. Smaller QDs emit shorter wavelengths (blue) and larger QDs emit longer wavelengths (red). This helps in precise control of colour output which makes them suitable to be used in display devices.

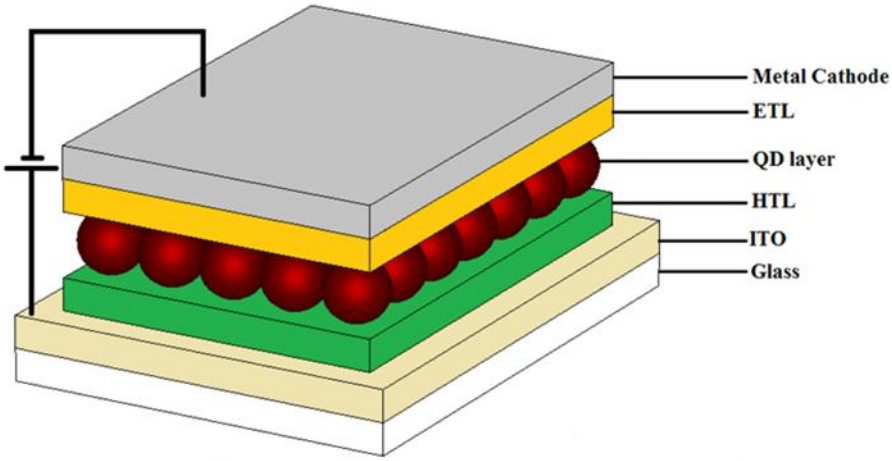


Fig. 12 QLED structure

Construction and Working: The structure of QLED is similar to the fundamental design of OLED (Fig. 12). However, unlike OLEDs which are self-emissive, QLEDs rely on backlight – typically a blue LED. In QLEDs, light emission is not due to organic or polymer materials, but rather from the excitation of quantum dots, such as cadmium selenide (CdSe) nanocrystals, which convert the backlight into highly pure red and green light. Ideally, the QD layer is placed between the backlight and LCD panel, sandwiched between the electron and hole transport layers. Electrons and holes injected from the electrodes move in to the QD layer, where they recombine. Depending on the size and material of the QDs, light of specific colours are emitted. The emitted light illuminates the LCD panel, which modulates brightness and image details using liquid crystal alignment. Although QLEDs are less efficient compared to OLEDs, it provides clear image quality with high brightness levels, making it ideal for large size displays.

Applications of QLEDs

1. **Televisions and displays**: QLED displays are commonly used in televisions, monitors, smartphones, and other electronic devices. They offer superior image quality and color accuracy compared to traditional LCD displays.
2. **Lighting**: QLEDs can also be used as a source of lighting in various applications, including automotive lighting, street lighting, and architectural lighting. They offer energy-efficient and highly customizable lighting options.
3. **Medical imaging**: QLEDs can be used in medical imaging applications, such as in MRI machines, to produce high-resolution and accurate images.
4. **Advertising displays**: QLED displays can be used in advertising displays, such as digital billboards and signage, to produce high-quality and eye-catching visuals.

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